

Automation Workflows for Economic Research with Claude Code

Some Use Cases

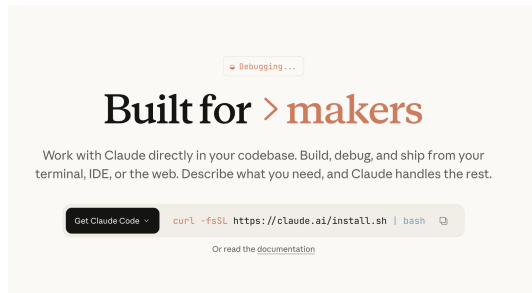
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Claude Code and "Vibe Coding": Democratizing AI-Assisted Research

What is Claude Code?

Claude Code is an AI agent released in February 2025 that executes natural-language descriptions to write, execute, and debug code autonomously.



Why This Matters for Economic Research

- ▶ **Efficiency:** Researchers can develop econometric tools, data pipelines, and visualizations faster than traditional coding
- ▶ **Accessibility:** Eliminates programming as a barrier to computational research

Source: Korinek, A. (2025). AI Agents for Economic Research.

Three Research Automation Use Cases

Use Case 1: Understand and Explore Replication Files

Use Case 2: Literature Organization and Web Scraping

Use Case 3: NLP Model Implementation

Use Case 1: Replication Package Exploration

Research Challenge

GitHub replication kits often lack comprehensive documentation. Exploration requires cloning repositories, parsing multiple file formats, generating summary statistics, creating visualizations, and documenting findings.

Claude Code Automation

- ▶ Clones the repo
 - ▶ Documents directory structure
 - ▶ Loads and summarizes all datasets
 - ▶ Generates 10-15 visualizations

Capabilities

- ▶ Autonomous repository cloning and structural mapping.
- ▶ Automated documentation and reporting.

Task1

great, now read /Users/rz/Desktop/AAU25/Test Claude Code/Task1.md and try to do it!

Task1.md#8-8

● *Thinking* ▾

● **Read** Task1.md

● *Thinking* ▾

● Perfect! I'll complete the KPSS Innovation Analysis task. Let me start by creating a plan and working through it systematically.

● **Update Todos**

- ☒ Create TASK1_KPSS_Innovation_Analysis folder structure
- ☐ Clone the KPSS2017 repository from GitHub
- ☐ Document directory structure and inventory all files
- ☐ Analyze and document all datasets (dimensions, variables, statistics)
- ☐ Create 10-15 publication-quality visualizations
- ☐ Generate comprehensive EXPLORATION_REPORT.md

Task1

Task 1 Complete!

I've successfully completed the KPSS Innovation Analysis task. Here's what was accomplished:

Project Location

[TASK1_KPSS_Innovation_Analysis/](#)

Deliverables Completed

1. Repository Cloned

- KPSS2017 Replication Kit (802 MB)
- 29 code files (Stata + Matlab)
- 8 datasets with 85+ million records

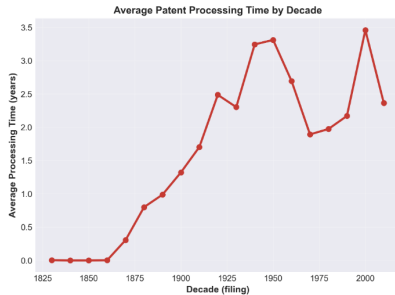
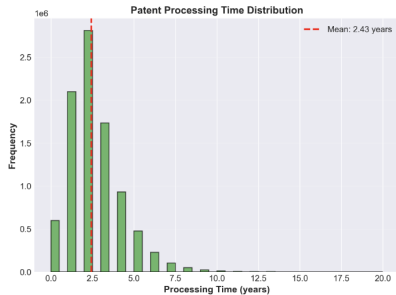
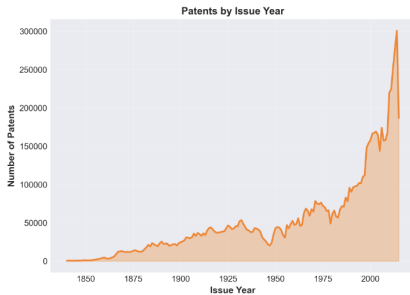
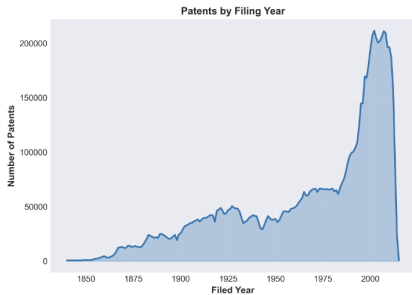
2. Directory Structure Documented

- Complete file inventory
- [DIRECTORY_STRUCTURE.md](#)

3. Data Analysis Completed

- **8 datasets** analyzed:
 - full_CPC_class: 33.1M rows
 - PatentDatesClasses: 9.1M rows
 - PatentChar: 9.1M rows
 - PatentSimilarity: 9.1M rows
 - Patent2CPCdummy2: 9.0M rows
 - Assignees: 7.3M rows
 - PatentCitations: 6.7M rows

Task1



Use Case 2: Literature Organization and Web Scraping

Research Challenge

Collecting papers from conference websites requires manual navigation, metadata extraction, PDF downloads, and standardized file organization. Error handling for broken links and paywalls adds additional complexity.

Claude Code Automation

```
Scrape https://www.remoteworkconference.org/  
Extract: Title, Authors, Affiliations, Abstract, DOI.  
Download PDFs (naming: LastName_Year_Title.pdf).  
Create an Excel metadata sheet with download status.
```

Capabilities

- ▶ HTML parsing and metadata extraction with error handling.
- ▶ Batch PDF downloads with retry logic for failed links.
- ▶ Automated file naming and folder organization.
- ▶ Excel workbook generation with comprehensive tracking.

Task2

read this task description and complete the task:/Users/rz/Desktop/AAU25/Test Claude Code/Task2.md

Task2.md#1-69

Thinking ~

I'll help you complete this literature review task. Let me start by creating a plan and then execute each step.

Update Todos

- ☒ Fetch conference website to analyze structure and available papers
- ☐ Create folder structure (main folder with /PDFs and /Metadata subfolders)
- ☐ Extract paper metadata from website (titles, authors, abstracts, etc.)
- ☐ Download all available PDF files
- ☐ Create Excel metadata tracking sheet with all required columns
- ☐ Generate completion summary report

Web Fetch <https://www.remoteworkconference.org/>

Fetches from <https://www.remoteworkconference.org/>

Thinking ~

Excellent! I've found the conference website and extracted information about all papers. This is the 2025 Remote Work Conference at Stanford University (October 22-24, 2025) with approximately 40+ papers.

Update Todos

- ☒ Fetch conference website to analyze structure and available papers
- ☒ Create folder structure (main folder with /PDFs and /Metadata subfolders)
- ☐ Extract paper metadata from website (titles, authors, abstracts, etc.)

Task2

Task Completed Successfully!

I've successfully completed the literature review task for the 2025 Remote Work Conference. Here's what was accomplished:

Summary Statistics

- **Total papers found:** 35
- **PDFs successfully downloaded:** 27 (77.1%)
- **Papers without accessible URLs:** 8

Deliverables Created

1. **Folder Structure** - [Conference_Papers_RemoteWork_2025/](#)
 - [/PDFs](#) - Contains 27 downloaded papers
 - [/Metadata](#) - Contains the Excel tracking sheet
2. **Excel Metadata File** - [Conference_Papers_Metadata.xlsx](#)
 - All 13 specified columns populated
 - Complete metadata for all 35 papers
 - Download status tracked for each paper
 - Auto-formatted for readability
3. **PDF Files** - All using the naming convention:
 - Format: `[LastName]_2025_[First_3_Title_Words].pdf`
 - Examples: `Zavarzin_2025_Work_from_Home.pdf`, `Harrington_2025_The_Power_of.pdf`
4. **Summary Report** - [TASK_COMPLETION_SUMMARY.md](#)
 - Detailed breakdown of all results
 - Session distribution analysis
 - Download failure reasons
 - Recommendations for obtaining missing papers

Task2

A	B	C	D	
Paper ID	Paper Title	Primary Author	All Authors	Author Address
1	Work from Home and Spatial Misallocation	Mikhail Zavarzin	Mikhail Zavarzin	
2	Remote Work and Consumer Cities	Yichen Su	Yichen Su; Franklin Qian	
3	Shopping from Home	Stephanie Johnson	Stephanie Johnson; Scott Baker; Nicholas Bloom; Jana Obradovic	
4	Do Firms Know what Workers Want?	Simon Cordes	Simon Cordes; Max Müller	University of Bonn
5	Internal Communication and Remote Work	Kyle Schirmann	Kyle Schirmann; Prithwiraj Choudhury; Miguel Espinosa; Tarun K	
6	So, dear applicant, do you mean working from home or shirking?	Elaine Moens	Elaine Moens; Elsy Verhofstadt; Luc Van Ootegem; Stijn Baert	
7	Wage Penalties for Workplace Amenities: Evidence from remote work	Jacob Sano	Jacob Sano	
8	Remote Work Makes Organizations Top-Heavy	Christopher Stanton	Christopher Stanton	
9	Organizational Technology Ladders: Remote Work and Generative AI	Gregor Schubert	Gregor Schubert	
10	Working from Home and Sorting of Male and Female Workers	Chiara Bernardi	Chiara Bernardi	Queen Mary University of London
11	Home Alone: Remote Work, Lonely Leisure and Mental Health	Natalia Emanuel	Natalia Emanuel; Emma Harrington; Amanda Pallais	
12	The Changing Nature of Work: What can we learn from Time Use	Francesca Foliano	Francesca Foliano; Rebecca Riley; Lynsey Brown	
13	The Daily Dynamics of Informal Interactions and Psychological Stress	Sumer Vaid	Sumer Vaid; Dylan Wiwad; Razan Sadaq-Keyes; Michael Massim	
14	The Power of Proximity to Coworkers	Emma Harrington	Emma Harrington; Natalia Emanuel; Amanda Pallais	
15	The Impact of Return-to-Office Mandates on Equity Analysts	Baolian Wang	Baolian Wang; Peixin Li; Jiawei Yu	
16	Work-From-Home and the Rug Rat Race: The New Geography of Remote Work	Francesco Saverio Lenzi	Francesco Saverio Lenzi	
17	The Gendered Effects of Work-from-Home on Spousal Labor Market Participation	Lily Seitelman	Lily Seitelman	
18	When Help Doesn't Help: Fathers' WFH During COVID and Its Labor Market Implications	Meng Su	Meng Su	London School of Economics
19	A Remote Work Baby Boom? Remote Work and Fertility in the United States	Siyao Lu	Siyao Lu; Emma Zang; Kelly Musick	
20	Work from Home and Fertility	Katelyn Cranney	Katelyn Cranney; Pablo Zarate; Cevat Giray Aksoy; Jose Maria B	
21	Childcare in the Era of Remote Work: Evidence from Child Care Tax Credits	Dmitri Koustas	Dmitri Koustas; Yana Gallen; Stephanie Karol; Ittai Lurie	
22	Flexible Work, Occupational Constraints, and the Dynamics of Female Labor Force Participation	Megha Patnaik	Megha Patnaik; Sara Casella; Kieran Larkin	
23	When Parents Work from Home	Michele Belot	Michele Belot; Pascal Achard; Arnaud Chevalier	

Task2

- ▼ TASK2_Conference_Papers_RemoteWork_2...
- ▼ Metadata
 -  ~Conference_Papers_Metadata.xlsx
 -  Conference_Papers_Metadata.xlsx
- ▼ PDFs
 -  Adharvyu_2025_Connecting_Rural_Wom...
 -  Aguilar_2025_The_Impact_of.pdf
 -  Belot_2025_When_Parents_Work.pdf
 -  Bernardi_2025_Working_from_Home.pdf
 -  Cranney_2025_Work_from_Home.pdf
 -  Escobar_2025_Remote_Work_and.pdf
 -  Fenizia_2025_Not_Incentivized_Yet.pdf
 -  Harrington_2025_The_Power_of.pdf
 -  Johnson_2025_Shopping_from_Home.pdf
 -  Koustas_2025_Childcare_in_the.pdf
 -  Lenzi_2025_Work_From_Home_and_the...
 -  Lu_2025_A_Remote_Work.pdf

Use Case 3: NLP Model Implementation

Research Challenge

Implementing transformer-based NLP pipelines requires environment configuration, model downloading, embedding generation, classifier training, and visualization. Iterative debugging cycles create substantial workflow friction.

Claude Code Automation

```
Create a Jupyter notebook implementing  
PatentSBERTa (AI-Growth-Lab/PatentSBERTa). Load 20  
patent claims, generate embeddings, train a KNN  
classifier, and visualize semantic similarity.
```

Capabilities

- ▶ Dependency management and environment setup.
- ▶ Model loading and embedding generation pipeline.
- ▶ Classifier training and hyperparameter configuration.
- ▶ Publication-quality visualization generation.
- ▶ End-to-end code testing and reproducibility verification.

Task3

now read /Users/rz/Desktop/AAU25/Test Claude Code/Task3.md and try to do it!

Task2.md

Thinking ▾

Read Task3.md

Thinking ▾

Perfect! I'll create a comprehensive PatentSBERTa implementation. Let me start by setting up the project structure and creating a Jupyter notebook.

Update Todos

- ☒ Create Task3_PatentSBERTa folder structure with notebooks/, data/, utils/
- ☐ Create sample patent claims dataset with IPC classes
- ☐ Build Jupyter notebook with PatentSBERTa implementation
- ☐ Implement embeddings generation and KNN classification
- ☐ Add visualization and similarity analysis
- ☐ Create utility functions and documentation

Task3

🔔 Task 3 Complete! ✓

I've successfully completed the **PatentSBERTa Implementation** task. Here's what was created:

📁 Project Location

[Task3_PatentSBERTa/](#)

✓ Deliverables Created

1. Jupyter Notebook - Main Implementation

- [PatentSBERTa_Demo.ipynb](#)
- **13 comprehensive sections:**
 1. Setup & imports
 2. Load PatentSBERTa model from HuggingFace
 3. Load patent claims dataset
 4. Generate embeddings
 5. Calculate similarity matrix
 6. Find similar patents
 7. Train KNN classifier
 8. Evaluate performance
 9. Visualize embedding space (t-SNE)
 10. Cross-class similarity analysis
 11. Distance analysis
 12. Summary & conclusions
 13. Export results

2. Sample Dataset - 20 Patent Claims

Task3

GenerateCodeMarkdownRun AllClear All OutputsOutline...

Select Kernel

PatentSBERTa: Deep NLP-Based Patent Classification

This notebook demonstrates the **PatentSBERTa** model, a hybrid approach combining:

- **Augmented SBERT embeddings** for semantic representation of patent claims
- **K-Nearest Neighbors (KNN)** for classification based on embedding similarity

Reference

Paper: Bekamiri et al. (2024), *PatentSBERTa: A deep NLP based hybrid model for patent distance and classification*

Model: [AI-Growth-Lab/PatentSBERTa](#) (HuggingFace)

Notebook Overview

1. **Setup:** Load libraries and PatentSBERTa model
2. **Data:** Load sample patent claims with IPC classifications
3. **Embeddings:** Generate semantic embeddings for patent texts
4. **Classification:** Train KNN classifier on embeddings
5. **Visualization:** Analyze similarity patterns and embedding space
6. **Results:** Evaluate classification performance

1. Setup and Imports

First, we import all necessary libraries and configure the environment.

```
# Standard library imports
import sys
import warnings
warnings.filterwarnings('ignore')
```

Integration Strategy and Best Practices

Stata MCP Integration Points

- ▶ **Use Case 1 (Replication Kit):** Heavy integration—summary statistics, econometric tables, and publication-grade figures via Stata.
- ▶ **Use Case 2 (Literature Organization):** Minimal integration—primarily uses Python web scraping tools.
- ▶ **Use Case 3 (NLP):** No integration—transformers and scikit-learn are the primary dependencies.

Recommended Workflow

- ▶ Econometric analysis and statistical inference → Stata MCP.
- ▶ Data wrangling and file organization → Claude Code.
- ▶ Machine learning and NLP pipelines → Claude Code.

Best Practices

- ▶ Review generated code before execution.
- ▶ Maintain version control for all agent-generated changes.
- ▶ Test on small samples before full-scale deployment.
- ▶ Reserve manual effort for novel methodology and statistical validation.